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Constitution and thermodynamics of Pt–La alloys

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Abstract

A complete set of thermodynamic functions was determined for the Pt–La system from electromotive force measurements on galvanic cells in the temperature range from 923 to 1073 K using CaF_2 single crystals as solid electrolytes. The phases on the platinum-rich side show pronounced negative deviations from ideal mixing behaviour; at 1073 K and infinite dilution, for instance, the relative partial excess Gibbs energy of La was determined to be −315 kJ/mol. Two effects are discussed to account for the deviations from ideal mixing behaviour: the transfer of the valence electrons of La to the electron gas of the alloy and the lattice distortion brought about by the size difference of the components. The phase relations were investigated by means of thermal, and X-ray analyses. The thermodynamic description of the system was optimized using experimental data from the present thermodynamic and phase diagram work. The calculated thermodynamic functions as well as the calculated phase diagram turned out to be in excellent accordance with the experimental results.
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1. Introduction

Thermodynamic data of alloys are not only of great practical importance but also very valuable in order to elucidate the factors which determine the thermodynamic stability of metallic phases. According to previous studies [1–5], the activities of a number of rare earth metals dissolved in Pd as the host metal exhibit pronounced negative deviations from Raoult's law. A bonding contribution resulting from the transfer of the valence electrons from the Fermi level of the electropositive component to the Fermi level of Pd was considered to be the responsible factor. Taking the lattice distortion brought about by size difference of the components as an additional effect, the thermodynamic behaviour of these alloys could be well described. These ideas should be also helpful to describe the thermodynamic properties of Pt–La alloys. One purpose of the present work was to provide the experimental information necessary to check these predictions.

Another purpose was to optimize the thermodynamic description of the system using both, thermodynamic and phase diagram data and to calculate the phase diagram. The experimental information available from the literature is scanty and does not allow a reliably construction of the Pt–La phase diagram. X-ray investigations have revealed the existence of six intermetallic compounds with the stoichiometries $LaPt_5$ [6,7], $LaPt_2$ [8–10], La_3Pt_4 [11], LaPt [12,13], La_3Pt_2 [14], and La_7Pt_3 [15,16]. Moreover, a eutectic was reported to exist on the Pt-rich side [17]. The tentative phase diagram given in a recent compilation [18] is based on these findings.

Thermodynamic results available from the literature include the enthalpies of formation of the compounds PtLa and Pt_2La [13,19,20], and the Gibbs energy of formation of the compound Pt_5La [21,22].

In the present work, the electromotive force (EMF) technique employing CaF_2 as a solid electrolyte was used to determine a complete set of thermodynamic functions for the Pt–La system. Reliable information on the phase relations was obtained by analyzing the concentration and temperature dependence of the activity. Moreover, the liquidus curve was fixed over the entire composition range by thermal analyses.

2. Experimental

2.1. Preparation of samples and materials used

Alloys were prepared from weighed amounts of Pt (99.9%; Degussa) and La (99.9%; Strem), both in the form of pieces, by induction melting on a water

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cooled copper boat under a protective atmosphere of carefully purified (see Chapter 2.2) Ar (99.998; Messer-Griesheim). Directly before weighing, a thin oxide layer was abraded from the surface of La with a tungsten carbide drill. The ingots were turned over and remelted several times in order to insure their homogeneity. The effect of magnetic stirring observed upon induction melting contributed to the homogenisation of the alloys. The resulting ingots were annealed in Ta crucibles for two weeks under purified Ar; alloys up to 70 at.% La were annealed at 1073 K, and those with higher La contents at 923 K. Compositions were checked by weighing; the maximal uncertainty in composition of about 0.2 at.% La was mainly caused by weight losses upon melting. Samples were reduced to powder by grinding with a tungsten carbide drill. The powder was then mixed with LaF_3 (99.9%; Chempur) in approximately 3 (metal) to 1 (fluoride) weight ratios and pressed into pellets to provide electrode material. CaF_2 single crystals (Korth) served as solid electrolytes.

2.2. Electromotive force measurements

Solid Pt–La alloys were studied over the entire range of composition on cells of the following type:

$$\text{Ta|La, LaF}_3\text{|CaF}_2\text{|Pt-La, LaF}_3\text{|Ta} \quad \text{(I)}$$

Even under the strongly reducing conditions established by the La, LaF_3 electrode, the CaF_2 electrolyte exhibits pure ionic conductivity [23]. Ta foils were used as electrical contacts in cell (I) which showed no reaction with the electrodes; they were connected to Pt leads. The electromotive force E of the symmetrical cell is directly related to the activity of La, a_{La}, by the expression

$$E = -\frac{RT}{3F}\ln a_{La} \quad (1)$$

A detailed description of the experimental arrangement is given in an earlier paper [23]. After the assembly of the cell, the apparatus was successively evacuated and refilled with Ar several times. The cell was then heated to 473 K and held at this temperature under vacuum for one hour. A slow flow of Ar was then adjusted which was maintained during the experiment. Oxygen contaminations of the electrodes have to be strictly avoided since they result in an erratic behaviour of fluorine concentration cells [23]; this calls for a careful purification of the protective gas. In order to remove traces of oxidizing impurities, bottled Ar was passed through a column with $Mg(ClO_4)_2$ at room temperature and over Zr turnings at 1273 K. The EMF was measured with a high impedance ($>10^9\ \Omega$) voltmeter (Keithley, model 199). The experiment was completely controlled by a computer, which helps to increase the number of data points in the temperature range of the measurements between 923 and 1073 K. After heating the cells up to 1073 K (923 K for alloys above 70 at.% La), a period of 12 h was allowed for the initial equilibration of the EMF. Subsequently, data points were taken at randomly selected temperatures to check the reversibility of the cells. Temperature and EMF were recorded after reaching constancy within ±1 K and ±0.3 mV, respectively. The observed absence of a hysteresis of the EMF indicates the thermal reversibility of the cells. As a further test, a small current (0.2–2.0 mA) was passed for a short time through the cell in either direction. After the imposed perturbation was removed, the EMF returned to its original value in a few minutes.

2.3. Differential thermal analysis

With the exception of the eutectic on the Pt-rich side, the liqidus curve and the temperatures of invariant reactions have not been established experimentally. In the present work, the phase transitions were investigated by DTA (Netsch DTA 404 S) on alloys with concentrations between 34 and 100 at.% La. To prevent a contamination of the samples, Al_2O_3 crucibles coated with La_2O_3 were used as sample holders. Au and Ag were chosen as reference materials. Purified Ar was passed through the apparatus to avoid sample oxidation. Since, the effect of undercooling was observed to occur, the measurements were made on heating with a heating rate of 10 K/min in general; control measurements with 5 K/min yielded identical results. The onset temperature was determined as the point of intersection of the extrapolated basis line with the straight line in the steepest part of the peak.

2.4. X-ray measurements

X-ray diffraction at room temperature was carried out with Cu Kα radiation using a goniometer (MZ VI, Seifert). The temperature was controlled to within ±0.1 K. In order to fix the limit of solid solubility of La in Pt, lattice parameters of pure Pt and of Pt-rich alloys were determined which had been annealed before for 24 h at 1073 K and then quenched to room temperature.

3. Results

3.1. EMF measurements

Forty alloys spanning the composition range of the system were investigated on cells of type (I) over varying temperatures between 923 and 1073 K. The temperature dependence of the cell potentials was essentially linear, and approximately 20 data points were collected from each run. The data from two runs on each alloy composition were treated by least square analysis to yield the results listed in Table 1.

Table 1
Results of the EMF measurements; $E(T) = E^\circ + (dE/dT)T$

at.% La	E° (V)	dE/dT (10^{-3} V/K)
2.0	1.8263	−0.4637
2.8	1.3810	−0.0345
4.7	1.5456	−0.2159
10.0	1.4736	−0.1193
14.3	1.7389	−0.3732
18.2	0.8943	0.0087
19.0	1.1552	−0.2367
19.0	1.1479	−0.2264
20.4	1.0551	−0.1435
21.9	0.6489	0.2049
22.8	0.8286	−0.0151
24.0	0.7988	0.0050
24.1	0.8282	−0.0086
24.9	0.6797	0.1087
26.0	0.8632	−0.0429
28.1	0.7941	0.0077
28.9	0.8424	−0.0335
29.8	0.7179	0.0487
30.6	0.8496	−0.0942
32.0	0.7951	−0.1627
35.0	0.3671	0.0310
36.1	0.3805	0.0302
38.8	0.3712	0.0326
41.4	0.3929	0.0060
44.8	0.3846	0.0178
46.0	0.3739	0.0233
47.0	0.3376	0.0537
49.1	0.3967	0.0039
52.4	−0.1830	0.2392
54.0	0.0315	0.0301
55.1	0.0513	0.0098
57.0	0.0445	0.0160
57.9	0.0725	−0.0075
58.1	−0.0152	0.0759
62.0	0.0840	−0.0199
63.2	0.0682	−0.0178
64.9	0.0494	−0.0117
68.3	0.0473	0.0022
85.0	0.0435	−0.0356
95.1	−0.0052	0.0060

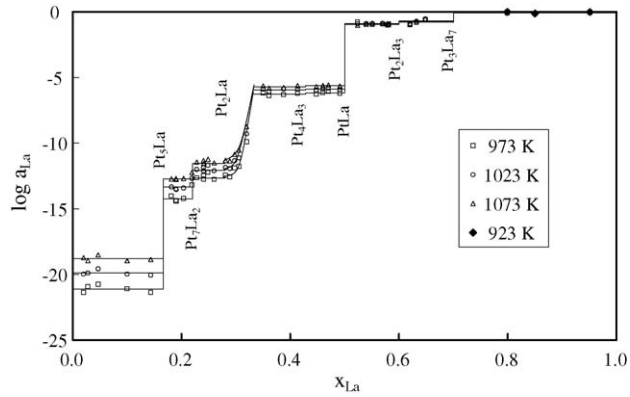

Fig. 1. Activity of La in Pt–La alloys (reference state: La solid).

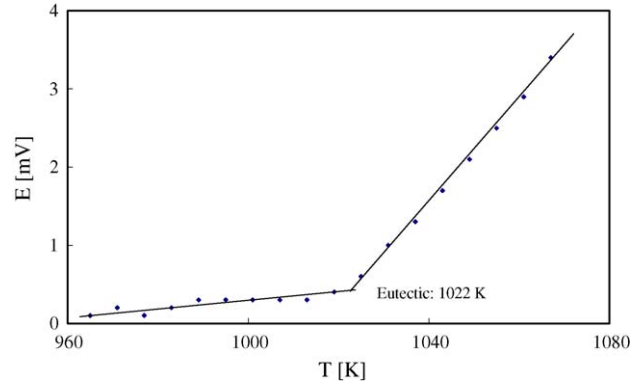

Fig. 2. EMF results measured on the cell La, $LaF_3|CaF_2|(La) + Pt_3La_7$.

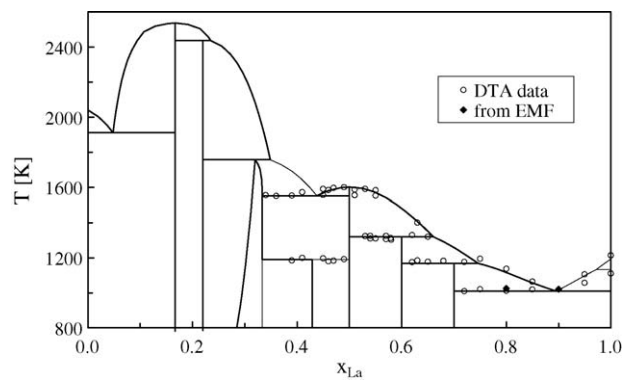

Fig. 3. Calculated phase diagram of the system Pt–La with experimental points.

The activities of La at 973, 1023, and 1073 K (923 K for alloys above 70 at.% La) calculated from Eq. (1) are shown on a logarithmic scale in Fig. 1. The concentration dependence of the activity isotherms indicates the existence of the six intermetallic compounds reported in the literature. In addition to these phases, the activity rise at 22 at.% La clearly proves the existence of a new compound. A comparison with analogous systems leads to the supposition that the stoichiometry of this compound is Pt_7La_2. With the exception of Pt_2La, the compounds are assumed to be line compounds.

The integral thermodynamic functions were calculated from the activity data by means of Gibbs-Duhem integrations. The free enthalpies ΔG at 1073 K and the enthalpies of formation ΔH, which are mean values calculated from the EMF data and their temperature dependence between 973 and 1073 K are listed in Table 2 in the columns ΔG (from EMF) and ΔH (from EMF).

The EMF versus T data in Fig. 2 were obtained by employing a two-phase mixture of (La) and Pt_3La_7 as the alloy electrode in cell (I). The data above and below 1022 K are straight lines with different slopes. The point of intersection is the eutectic temperature, which can be obtained from this plot with high precision. Phase transition data determined from EMF versus T plots are shown in Fig. 3 along with the DTA results.

3.2. Differential thermal analysis

The phase transition data are listed in Table 3 and are graphically shown in Fig. 3 along with the data determined from EMF. The reproducibility of the DTA results was approximately ±15 K. This relatively high inaccuracy is mainly attributed to the extremely strong affinity of La to oxygen. The experimental transition temperatures listed in Table 6 are mean values calculated from the data in Table 3 and they have a lower error limit accordingly.

These results helped to optimize the thermodynamic date sets and to establish the phase diagram.

Table 2
Integral thermodynamic properties of Pt–La intermetallic compounds: ΔG at 1073 K and mean values for ΔH in the range 973–1073 K

Phase	at.% La	ΔG (from EMF) (kJ/mol)	ΔG (calculated) (kJ/mol)	ΔH (from EMF) (kJ/mol)	ΔH (calculated) (kJ/mol)
Pt_5La	16.7	−64.4	−64.7	−76.9	−71.7
Pt_7La_2	22.0	−77.0	−77.3	−91.5	−84.3
Pt_2La	28.1	−88.1		−104.5	
	28.9	−89.7	−90.1	−106.0	−104.7
	29.8	−91.4	−91.6	−107.5	−107.1
	30.6	−92.9	−93.0	−108.9	−107.9
	32.0	−95.0	−94.9	−111.5	−109.8
	33.3	−96.0	−96.4	−112.6	−111.4
Pt_4La_3	42.9	−99.1	−99.2	−112.1	−112.3
PtLa	50.0	−101.1	−101.0	−111.8	−110.9
Pt_2La_3	60.0	−84.6	−85.9	−91.6	−87.0
Pt_3La_7	70.0	−67.0	−67.5	−73.2	−65.6

Table 3
DTA results measured on heating (standard: Ag or Au)

at.% La	DTA effects (K)
34.0	1555
36.0	1550
39.0	1185, 1553
41.0	1200, 1572
45.0	1197, 1556, 1590
46.0	1180, 1584
47.0	1184, 1596
49.0	1192, 1600
51.0	1587
51.0	1555
53.0	1324, 1591
54.0	1310
54.0	1325
55.0	1310, 1553
55.0	1553
57.0	1307
57.0	1325
58.0	1303
62.0	1175, 1330
63.0	1185, 1400
65.0	1177, 1320
68.0	1183
72.0	1010, 1176
75.1	1020, 1195
80.0	1012, 1137
85.0	1019, 1064
95.0	1057, 1106
100.0	1110, 1215

3.3. X-ray measurements

From 12 measurements on four (Pt) + Pt_5La two phase alloys which were reproducible to within ±0.01 pm, the lattice parameter at the boundary of (Pt) was determined to be 392.39 pm. In the small concentration range of (Pt), the lattice parameters a will follow Vegard's law ($a = a_{Pt} + bx_{La}$), and the knowledge of the slope b makes it possible to fix the terminal solid solubility of La in Pt. The slope b is related to the partial molar volume of La at infinite dilution by means of the expression

$$\bar{V}_{La}(x_{La} \to 0) = \frac{1}{4}N_A a_{Pt}^3 + \frac{3}{4}N_A a_{Pt}^2 b \quad (2)$$

where N_A is Avogadro's constant. Since the partial molar volumes of a number of metals show only slight differences when dissolved in Pd and Pt as host components, the partial molar volume of La in Pd $(18.2 \pm 1)\,cm^3/mol$ [24] is considered to be a good estimate for the corresponding quantity of La in Pt. From 10 measurements on pure Pt, which were reproducible to within ±0.01 pm, the lattice constant of Pt was determined to be 392.35 pm.

With these data the following expression was obtained for the lattice extension of (Pt) alloys:

$$a = 392.35 + 128x_{La}\ [\text{pm}] \quad (3)$$

Taking the lattice constant at the solubility limit into account, the solid solubility of La in Pt at 1073 K was estimated to be (0.03 ± 0.01) at.% La.

4. Optimization

The evaluation of the phase diagram was carried out with the computer program CHEMSAGE. This program package includes an optimization routine [25], which makes it possible to optimize the Gibbs energies of formation of the phases of the system by means of the least squares method [26] using simultaneously thermodynamic and phase diagram data. In the present optimization, the Gibbs energies of formation of the phases listed in Table 2, the DTA results listed in Table 3, and the eutectic on the Pt-rich side at $x_{La} = 0.05$ and $T = 1913 \pm 15$ K [17] were used as experimental input data. The result of the optimization yields a polynomial description of the Gibbs energies of all phases in the Pt–La system.

4.1. Thermodynamic modelling and description

The temperature dependence of the Gibbs energies of the pure stable elements, referred to the standard state (stable element reference, SER), are described by the formula

$$G_\Phi^\circ - H_\Phi^{SER} = A_\Phi + B_\Phi T + C_\Phi T \ln T + D_\Phi T^2 + E_\Phi T^3 + F_\Phi/T \quad (4)$$

The coefficients A_Φ to F_Φ are listed in Table 4. All values are taken from the SGTE data file [27]. There are different sets of coefficients for the same element in different temperature ranges.

All solid intermetallic phases with the exception of Pt_2La are treated as stoichiometric line compounds. Since the solid solubility of La in (Pt) and vice versa is extremely small, the data for the pure elements were used in the calculation. The temperature dependence of all these phases is described by Eq. (4) using the coefficients A_Φ to D_Φ.

The Redlich–Kister formula [28] was used to describe the concentration and temperature dependence of the excess Gibbs energy of the liquid phase, which results in the following expression

$$G_{liq} = \sum_{i=1}^{2} x_i (G_i^\circ - H_i^{SER}) + RT \sum_{i=1}^{2} x_i \ln x_i + G_{liq}^E,$$

$$G_{liq}^E = x_1 x_2 \sum_{k=0}^{n} (A_k + B_k T)(x_1 - x_2)^k \quad (5)$$

where A_k and B_k are adjustable coefficients.

According to the results of X-ray measurements [29], the homogeneity range of the phase Pt_2La extends at 1073 K from $x_{La} = 0.28$ up to the stoichiometric composition. A sublattice model based on the regular solution approximation [30] was used to describe the Gibbs energy over the phase field of Pt_2La. At the stoichiometric composition, Pt and La atoms occupy different lattice positions. Considering the fact that La is considerably larger than Pt and considering the occurrence of the Pt surplus in the compound, only the sublattice of La is assumed to dissolve both elements. The "limiting structures" of the solution sublattice are a complete occupation either with La or with Pt which may be described by $(Pt)_{2/3}\ (Pt)_{1/3}$ and $(Pt)_{2/3}\ (La)_{1/3}$; the

Table 4
Gibbs energies of the pure elements in J/mol [27]
$G_\phi^\circ - H_\phi^{SER} = A_\phi + B_\phi T + C_\phi T \ln T + D_\phi T^2 + E_\phi T^3 + F_\phi / T$

Element	T_{max}[a] (K)	A_ϕ	B_ϕ	C_ϕ	D_ϕ	E_ϕ ($\times 10^{-6}$)	F_ϕ
Pure solid substances							
Pt (fcc)	1300.00	−7595.631	124.38827	−24.5526	−0.002483	−0.020138	7974.00
	2041.50	−9253.174	161.52962	−30.2527	0.002322	−0.656946	−272106.00
	4000.00	−222048.22	1019.3589	−136.193	0.020455	−0.759259	71539020.00
La (dhcp)	550.00	−7968.403	120.2846	−26.34	−0.001295	0.00	0.00
	2000.00	−3381.413	59.06113	−17.165941	−0.008372	0.689320	−399448.00
	4000.00	−15608.882	181.39007	−34.3088	0.00	0.00	0.00
La (fcc)	1134.00	−6109.797	89.878761	−21.7919	−0.004045	−0.525865	0.00
	2000.00	−124598.98	955.87838	−139.34674	0.042032	−3.066199	20994153.00
	4000.00	−12599.386	178.54399	−34.3088	0.00	0.00	0.00
La (bcc)	800.00	−3952.161	88.072353	−21.7919	−0.004045	−0.525865	0.00
	1134.00	321682.67	−3565.0825	513.44071	−0.387295	49.54798	−36581228.00
	1193.00	−16377.894	218.49299	−39.5388	0.00	0.00	0.00
	2000.00	−136609.91	1123.344	−163.41307	0.053969	−4.056395	21167204.00
	4000.00	−8205.988	174.83632	−34.3088	0.00	0.00	0.00
Melts							
La	1134.00	5332.653	18.23012	−11.018819	−2.02E−02	2.937750	−133541.00
	4000.00	−3942.004	171.01843	−34.3088	0.00	0.00	0.00
Pt	600.00	12518.385	115.11309	−24.5526	−0.002483	−0.020138	7974.00
	2041.50	19023.49	32.94182	−12.340377	−0.011552	0.931516	−601426.00
	4000.00	1404.468	205.85896	−36.50	0.00	0.00	0.00

[a] Parameters valid up to T_{max}.

corresponding SGTE description is Pt:Pt and Pt:La. With these assumptions, the Gibbs energy of Pt_2La is given by [30]:

$$G_m = y_{Pt} G^\circ_{Pt:Pt} + y_{La} G^\circ_{Pt:La} + \frac{1}{3} RT(y_{La} \ln y_{La} + y_{Pt} \ln y_{Pt}) + y_{Pt} y_{La} \sum_k L^{(k)}_{Pt:Pt,La} (y_{Pt} - y_{La})^2,$$

$$\text{with } L^k_{Pt:Pt,La} = A_k + B_k T \quad (6)$$

where y_{Pt} and y_{La} are the mol fractions of the components on the solution sublattice.

The third term in Eq. (6) describes the excess Gibbs energy and is based on the Redlich–Kister formula with coefficients $L^{(k)}_{Pt:Pt,La}$ taken as linear functions of temperature.

4.2. Selection of adjustable coefficients

An important step in the optimization procedure is the selection of which and how many of the adjustable coefficients shall be used for the evaluation. By using too many coefficients, the resulting functions may follow the scatter of the experimental data and thus lead to unexpected curvatures. An insufficient number of coefficients, however, may result in an inaccurate description of the thermodynamic functions.

For the solid intermetallic phases the coefficients A_Φ and B_Φ were optimized, whereas, the coefficients C_Φ and D_Φ were estimated by Neumann–Kopp's rule.

4.3. Optimization results

The resulting set of coefficients valid for the temperature range 600–2000 K is listed in Table 5. The calculated phase diagram is shown in Fig. 3 along with the experimental data. Calculated and measured characteristic phase equilibria are listed in Table 6 for comparison. The calculated and experimental free enthalpies of formation of the solid intermetallic phases are listed in Table 2. Each of these comparisons indicates an excellent agreement between measured and calculated data.

5. Discussion

The most noteworthy characteristics of the thermodynamic properties of the Pt–La system are the extremely large negative deviations from ideality on the Pt-rich side of the system as indicated by the isotherms in Fig. 1. From the activity data in the (Pt) + Pt_5La two phase field the relative partial excess free energy

$$\Delta \bar{G}^E_{La} = RT(\ln a_{La} - \ln x_{La}) \quad (7)$$

at the solubility limit of La in (Pt) (0.03 at.% La) was determined to be −315 kJ/mol at 1073 K; because of the small solid solubility, this value may also be used for La dissolved at infinite dilution.

The exceptionally high thermodynamic stability of the Pt-rich solid solution may be tied to the electronic structure of the components. Measurements of the paramagnetic susceptibility on a number of Pt alloys [31] were interpreted in terms of a charge transfer model in which the valence electrons of the alloying partners of Pt are transferred to vacant 5d-states with a consequent reduction in paramagnetic susceptibility. A charge transfer from the Fermi level of La to the Fermi level of Pt gives rise to a bonding or electronic (superscript: e) contribution $\Delta \bar{G}^{Ee}_{La}$ to the excess function which dominates the mixing behaviour.

A quantitative thermodynamic analysis of the (Pt) solid solution has to include the influence of the lattice distortion brought about by the appreciable size difference of the components. The

Table 5
Set of calculated coefficients for the Pt–La system

	k	A_k	B_k
Redlich–Kister model: $G^E_{liq} = x_{Pt}x_{La}\sum_{k=0}^{1}[(A_k + B_kT)(x_{Pt} - x_{La})^k]$ (J/mol)			
Liquid phase	0	−314162.1	−22.569
	1	279334.7	−162.21

Phase	T_{max} (K)	A_ϕ	B_ϕ	C_ϕ	D_ϕ
Stoichiometric model: $G_\phi = A_\phi + B_\phi T + C_\phi T \ln T + D_\phi T^2$					
Pt_3La_7	2500	−76752.02	153.3909	−30.30	−0.0012
Pt_2La_3	2500	−97329.73	148.3840	−29.00	−0.0016
PtLa	2500	−120435.7	148.7557	−27.70	−0.0020
Pt_4La_3	2500	−121299.2	145.8077	−26.70	−0.0023
Pt_7La_2	2500	−91551.3	122.5498	−23.80	−0.0032
	4000	−133203.7	204.0897	−30.87	−0.0070
Pt_5La	2500	−78694.6	119.9326	−23.30	−0.0034
	4000	−271891.9	273.9334	−24.39	−0.0307
$G^\circ_{Pt:Pt}$ and $G^\circ_{Pt:La}$ (J/mol) for the description of the phase Pt_2La					
Pt:Pt	4000	−33474.33	326.75	−21.00	−0.0041
Pt:La	4000	−119703.38	140.516	−25.50	−0.0027

	k	A_k	B_k
$L^k_{Pt:Pt,La} = A_k + B_kT$ (J/mol) for the description of the phase Pt_2La			
Pt_2La	0	−21380.618	−320.20506
	1	25868.66	−115.46198

lattice distortion reduces the solid solubility and gives rise to an elastic or dilatational (superscript: d) contribution $\Delta\bar{G}^{Ed}_{La}$. Neglecting an interrelation between both effects, the relative partial excess free energy can be separated into electronic and elastic parts:

$$\Delta\bar{G}^E_{La} = \Delta\bar{G}^{Ee}_{La} + \Delta\bar{G}^{Ed}_{La} \quad (8)$$

As in previous publications [32,33] the following equation was used to calculate the elastic part of the excess function independent from thermodynamic information:

$$\Delta\bar{G}^{Ed}_{La}(x_{La} = 0) = \frac{(\bar{V}_{La} - V_{Pt})}{b_{Pt}} - \frac{V_{Pt}B_{Pt}}{b_{Pt}(1 - b_{Pt})} \times \left[1 - \left(\frac{\bar{V}_{La}}{V_{Pt}}\right)^{(1-b_{Pt})}\right] \quad (9)$$

In this expression which is based on a model of continuum elasticity, $\bar{V}_{La}$ is the partial molar volume of La at infinite dilution, V_{Pt} the molar volume of Pt, B_{Pt} the bulk modulus of Pt at $p = 1$ bar, and $b_{Pt} = (\partial B_{Pt}/\partial p)_T$. The $\Delta\bar{G}^{Ed}_{La}$ values calculated from Eq. (9) with $\bar{V}_{La} = 18.2\,cm^3/mol$ and the literature data $B_{Pt} = 2.76$ Pa and $b_{Pt} = 13.7$ [34] are listed in Table 7 along with the experimentally determined value $\Delta\bar{G}^E_{La}$ and its electronic part $\Delta\bar{G}^{Ee}_{La}$ obtained from Eq. (8).

The strong influence of lattice distortion as indicated by the large positive value of $\Delta\bar{G}^{Ed}_{La}$ is considered to be responsible for the extremely low solid solubility of La in Pt. The reduction of the misfit by a structural change favours the formation of the compound Pt_5 La, which is in equilibrium with the solid solution. The large misfit energy, however, is quite overwhelmed by the bonding contribution $\Delta\bar{G}^{Ee}_{La}$, which results in a pronounced negative value for $\Delta\bar{G}^E_{La}$.

The results of previous investigations on Pt–Zr [34] and Pt–Al [35] alloys were analyzed in an analogous way; the results are included in Table 7 for comparison. In contrast to La, the partial molar volumes of Zr and Al are close to the molar volume of Pt. Therefore, the elastic parts of the excess functions of Zr and Al are distinctly smaller compared to the corresponding value of La. Because of the distinctly smaller influence of the misfit, the solid solubility of Zr and Al in Pt extends up to about 20 at.% Zr and Al, respectively. The mixing behaviour of Zr and Al in Pt is overwhelmingly governed by the electronic contributions.

Table 6
Phase equilibria in the Pt–La system

Reaction	Type of reaction	T calculated (K)	T experimental (K)
L ↔ La(fcc) + Pt_3La_7	Eutectic	1009	1015 ± 10
Pt_3La_7 ↔ L + Pt_2La_3	Peritectic	1167	1179 ± 10
Pt_2La_3 ↔ L + PtLa	Peritectic	1319	1317 ± 10
Pt_4La_3 ↔ Pt_2La + PtLa	Peritectoid	1190	1190 ± 10
L ↔ Pt_2La + PtLa	Eutectic	1552	1557 ± 10
Pt_2La ↔ L + Pt_7La_2	Peritectic	1760	
Pt_7La_2 ↔ L + Pt_5La	Peritectic	2440	
L ↔ (Pt) + Pt_5La	Eutectic	1911	1913 ± 15 [17]

Table 7
Excess potentials with elastic and electronic parts at 1073 K (values refer to infinite dilution)

System	$\Delta\bar{G}^E_i$ (kJ/mol)	$\Delta\bar{G}^{Ed}_i$ (kJ/mol)	$\Delta\bar{G}^{Ee}_i$ (kJ/mol)
Pt–La	−315 ± 6	167 ± 40	−482 ± 46
Pt–Zr [34]	−362 ± 4	47 ± 9	−409 ± 13
Pt–Al [35]	−245 ± 4	27 ± 5	−272 ± 9

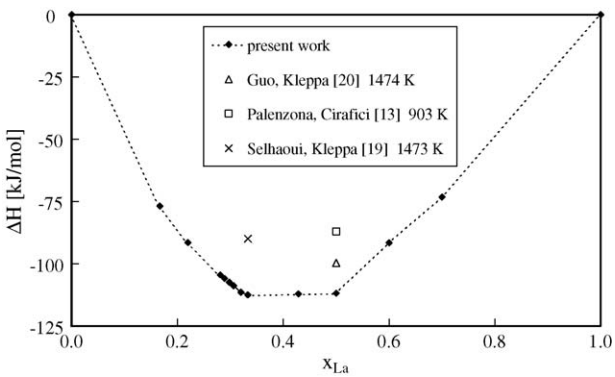

Fig. 4. Enthalpies of formation of Pt–La alloys.

The extremely high thermodynamic stability of the compounds in the Pt–La system as indicated by the pronounced negative values of ΔG and ΔH (see Table 2) may be also attributed to the transfer of the valence electrons of La to the electron gas of the inter-metallic compounds.

The heats of formation obtained from the ΔG data between 973 and 1073 K are shown in Fig. 4 along with calorimetrically determined data of the compounds Pt_2La and PtAl. The comparison in Table 2 illustrates the excellent agreement between ΔH values from EMF and from the optimization calculation. Compared to these results, the calorimetrically determined data are less exothermic. The same trend, ΔH from EMF more exothermic then ΔH determined calorimetrically was also observed in previous investigations on Pt–Y [29], Ni–La [36], and Ni–Gd [37] alloys.

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